

Surface passivation of perovskite film for efficient solar cells

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In recent years, the power conversion efficiency of perovskite solar cells has increased to reach over 20%. Finding an effective means of defect passivation is thought to be a promising route for bringing further increases in the power conversion efficiency and the open-circuit voltage (V_{OC}) of perovskite solar cells. Here, we report the use of an organic halide salt phenethylammonium iodide (PEAI) on $\text{HC}(\text{NH}_2)_2\text{-CH}_3\text{NH}_3$ mixed perovskite films for surface defect passivation. We find that PEAi can form on the perovskite surface and results in higher-efficiency cells by reducing the defects and suppressing non-radiative recombination. As a result, planar perovskite solar cells with a certificated efficiency of 23.32% (quasi-steady state) are obtained. In addition, a V_{OC} as high as 1.18 V is achieved at the absorption threshold of 1.53 eV, which is 94.4% of the Shockley–Queisser limit V_{OC} (1.25 V).

The power conversion efficiency (PCE) of perovskite solar cells (PSCs) swiftly increased from 3.8% to more than 20% in 10 years due to composition engineering, perovskite film growth control and perovskite/transport layer interface engineering^{1–13}. To further increase the PCE of PSCs, improvement of the open-circuit voltage (V_{OC}) and fill factor (FF) could be more feasible than the short-circuit current (J_{SC}), considering that the J_{SC} is almost close to its limit ($\sim 26 \text{ mA cm}^{-2}$) for a bandgap of around 1.55 eV.

It is known that the V_{OC} of solar cells is strongly related to the recombination of defects in the bulk or on the surface of the absorber layer^{14–17}. Solution-processed perovskite films are usually polycrystalline, which means they contain substantial structural disorder, such as grain boundary defects and crystallographic defects^{14–17}. Although it has been theoretically predicted that most of the defects formed in the perovskite layer are shallow defects¹⁸, several experimental results have shown that the suppression of defects in solution-processed perovskite films is critical for further enhancing the performance of PSCs towards their thermodynamic limits^{9,19,20}.

The surface is the place where defects are most easily formed, and passivation of the surface defects is always the most important task in any type of solar cells^{11,15,21}. In PSCs, several efficient surface passivation methods have been adopted. For example, excess PbI_2 in the perovskite layer, on the surface or at the grain boundary can suppress charge recombination by formation of I-type band alignment^{22–24}. Wide bandgap perovskites such as $\text{FAPbI}_{3-x}\text{Br}_x$ (ref. 25), two-dimensional (2D) perovskite materials, for example, PEA_2PbI_4 and BA_2PbI_4 (refs. 26–31), and super-thin insulated materials such as polystyrene¹⁵ have been utilized. More recently, several halide compounds such as quaternary ammonium halides¹¹, diammonium iodide²⁰, potassium halide³² and iodopentafluorobenzene³³ have been employed to passivate dangling bonds in the perovskite layer by exploiting the associated intermolecular interaction.

Here, we develop an organic halide salt, phenethylammonium iodide (PEAI), for post-treatment of mixed perovskites $\text{FA}_{1-x}\text{MA}_x\text{PbI}_3$ (FA: $\text{HC}(\text{NH}_2)_2$; MA: CH_3NH_3) to suppress the surface defects of perovskite polycrystalline films for efficient solar cells. Surprisingly, the organic halide salt PEAi, rather than the 2D

layered PEA_2PbI_4 perovskite proposed in many recent studies^{26–31}, serves as a much more effective passivation additive for a 3D perovskite. We carefully control the conversion process of PEAi to PEA_2PbI_4 in the actual devices using direct X-ray diffraction (XRD) evidence and show that the existence of PEAi instead of PEA_2PbI_4 in the perovskite thin-film devices is the key to increasing the V_{OC} to as high as 1.18 V, which is 94.4% of the Shockley–Queisser limit V_{OC} (1.25 V). As a result, a 23.32% certificated PSC has been achieved.

Design and fabrication of the perovskite films

The perovskite layer $\text{FA}_{1-x}\text{MA}_x\text{PbI}_3$ was deposited using a two-step spin-coating method^{8,24,34,35}, and the mixed organic salts FAI and MAI were diffused into the PbI_2 bottom layer via annealing. Comparing with previous reports and ours^{5,8,24}, we replaced the methylammonium bromide (MABr) in the traditional high-performance perovskite material FAPbI_3 (MAPbBr_3) with methylammonium iodine (MAI) to form $\text{FA}_{1-x}\text{MA}_x\text{PbI}_3$ perovskites, which is expected to further lower the absorption threshold of the perovskites to absorb more sunlight (Supplementary Fig. 1). We found that a small amount of MAI incorporation (8% in precursor) can stabilize the α -phase of FAPbI_3 (Fig. 1c), which is consistent with the tolerance factor empirical rule^{36,37}. To introduce a passivation layer, a PEAi salt solution was spin-coated onto the perovskite surface. It is noted that no additional process was carried out for PEAi layer. The device structure of the perovskite solar cells we adopted in this study is shown in Fig. 1a. We infer that the PEAi coating on the perovskite layer fills the iodine vacancies on the surface (Fig. 1b), and at the grain boundary (Supplementary Fig. 2), resulting in passivation of the surface defects and enhancement of the device performance²⁰.

Characterization of the perovskite films

We carried out grazing incident X-ray diffraction (GIXRD) for the perovskite films with and without a PEAi layer (Fig. 1c) to clearly show the states of PEAi on the surface. We found that the diffraction peaks for the perovskite with PEAi treatment are almost the same as for the control film without PEAi. However, a new diffraction

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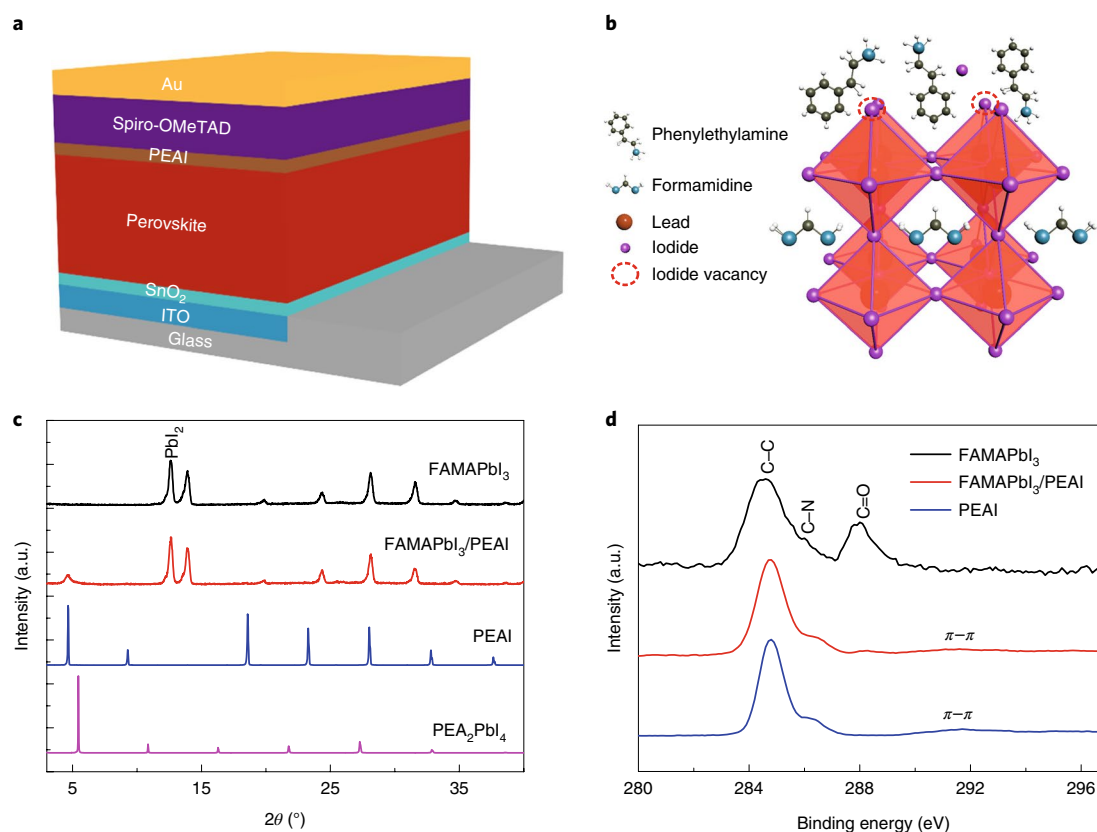


Fig. 1 | Device structure, possible passivation mechanism and states of PEAI on the perovskite surface. a, The device structure adopted in this study. PEAI is used for post-treatment of the perovskite surface. **b**, Possible passivation mechanism of the PEAI layer for the perovskite film. **c**, GIXRD patterns of perovskite films before and after PEAI treatment. The diffraction patterns of the PEAI powder and of the PEA_2PbI_4 film are also shown. **d**, XPS spectra of the perovskite before and after PEAI treatment, and of the pure PEAI film. The C 1s core-level energy spectra are shown for comparison.

peak appeared at 4.7° , which is consistent with the diffraction pattern for PEAI powder, and there is no obvious diffraction peak at 5.4° corresponding to the 2D perovskite PEA_2PbI_4 (Fig. 1c and Supplementary Fig. 3). It can be concluded that the dominant form of PEAI salt on the perovskite surface is the PEAI crystal itself, which is in contrast to previous reports on the formation of a 2D perovskite on a 3D perovskite surface^{26–31}. We will show that PEAI itself has a much better passivation effect than a 2D PEA_2PbI_4 thin layer. From scanning electron microscopy results (Supplementary Fig. 2a,b), we found that the perovskite surface was covered by a PEAI layer. In addition, the roughness of the perovskite layer was reduced from 32 to 24 nm according to atomic force microscopy results (Supplementary Fig. 2c,d), indicating that the PEAI salt is more easily deposited at the grain boundary (usually, the grain boundary is the deepest place on the surface, Supplementary Fig. 7). Therefore, this reduces the height difference between the grain surface and the grain boundary.

We conducted X-ray photoelectron spectroscopy (XPS) measurements to confirm the existence of PEAI on perovskite (Fig. 1d). For the perovskite with PEAI treatment, in addition to C–C and C–N peaks located at 284.6 eV and 286.1 eV, respectively, we also found that a peak emerges at a binding energy of 292.0 eV, consistent with π – π bonding in the phenyl functional group of the PEA^+ cations²⁶. We also noticed that the C=O peak (288.1 eV) seen for the control film associated with oxygen/moisture is significantly suppressed after PEAI modification, indicating that the PEAI coating can slow the degradation of the perovskite layer. Furthermore, according to the Pb 4f and I 3d core-level energy spectra (Supplementary Fig. 4), it can be estimated that the Pb:I ratio for the control perovskite film

is 1:1.55, which indicates that the surface is seriously iodine deficient, which could lead to obvious point defects (Fig. 1b). After PEAI modification, the Pb:I ratio dramatically increased to 1:2.34, indicating the presence of abundant iodide on the perovskite surface and that the iodine vacancy is likely to be filled (Fig. 1b).

We monitored the steady photoluminescence (PL) of the perovskite films with PEAI treatment under different conditions (Fig. 2a). We found that the PL intensities are obviously increased after PEAI treatment without additional thermal annealing, indicating that recombination in the perovskite layer has been largely suppressed^{38,39}. We also mapped the PL intensities of perovskite films with 470 nm excitation (Supplementary Fig. 5), and found that the PL from the perovskite films with PEAI treatment becomes much more intense and uniform compared with the control film. Owing to the strong absorption of perovskite at short wavelengths, we anticipate that only the perovskite surface (<100 nm) is excited when using 470 nm excitation. Therefore, strong PL enhancement indicates that PEAI treatment suppresses the surface non-radiative recombination defects. According to the time-resolved PL results (Fig. 2b, Supplementary Fig. 6 and Supplementary Table 1), the carrier lifetime of the perovskite is significantly increased from 0.3 μs to more than 2 μs , and the recombination rate has been reduced by fivefold after PEAI treatment.

It is interesting that we found that the PL intensity and lifetime of the perovskite films become slightly weaker and shorter when the PEAI layer is annealed above 50°C (Fig. 2a,b). This phenomenon becomes more obvious for thermal annealing of the perovskite/PEAI film at 100°C (Fig. 2a,b), and the PEAI-treated perovskite film shows almost the same emission properties as the control film,

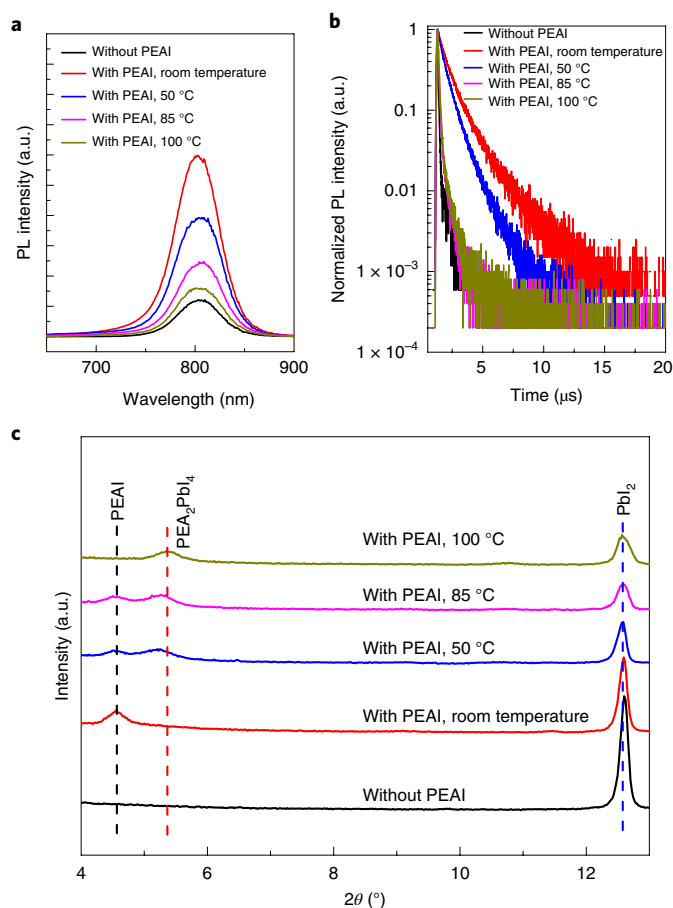


Fig. 2 | Surface passivation of perovskite layer by PEAI. **a**, Steady PL of the perovskite films with PEAI treatment under different conditions. **b**, Time-resolved PL of the perovskite films with PEAI treatment under different conditions. **c**, XRD of perovskite films with PEAI treatment under different conditions.

indicating that the passivation effect has almost disappeared. From the XRD results (Fig. 2c), we found that the PEAI is gradually converted into the PEA_2PbI_4 phase during the thermal annealing process, and when the perovskite/PEAI layer is annealed at 100 °C for 5 min, the PEAI crystal diffraction totally disappears (Fig. 2c). According to the PL and XRD results (Fig. 2), it can be concluded that the PEAI itself shows a much better passivation effect than the 2D perovskite of PEA_2PbI_4 . Therefore, in this study, we utilized the PEAI layer, not the conventional 2D perovskite^{26–31}, to passivate the defects in the perovskite layer.

Device fabrication and characterization

Keeping in mind that PEAI can efficiently passivate the surface defects in the perovskite layer, we fabricated devices with the structure ITO/SnO₂/perovskite/spiro-OMeTAD/Au^{8,24} (ITO: indium tin oxide; spiro-OMeTAD: 2,2',7,7'-tetrakis(*N,N*-dimethoxyphenylamine)-9,9'-spirobifluorene). The thicknesses of the electron transport layer SnO₂, the perovskite layer and the hole transport layer spiro-OMeTAD were about 40 nm, 800 nm and 250 nm, respectively (Supplementary Fig. 7). The dependence of the device performance on the PEAI concentration (Fig. 3a, Supplementary Fig. 8 and Supplementary Table 2) shows that the PCE jumped significantly when the perovskite films were treated with a low concentration of PEAI solution, and the PCE reached a maximum at 20 mM of PEAI. For this best condition, the mass

thickness of the PEAI is about 16 nm when deposited on a smooth silicon substrate (Supplementary Fig. 9). It is noted that several tens of nanometres of PEAI films deposited on a very rough perovskite surface (roughness = 32 nm, Supplementary Fig. 2) via a solution process cannot be conformal, and the resulting PEAI film should be discontinuous or thin enough for charge transport (Supplementary Fig. 10). While the device performance began to drop at high PEAI loading even the recombination rate was low (Supplementary Fig. 6), which could be due to the highly insulating properties of PEAI. The typical *J*–*V* curve of the device with PEAI (20 mM) and without PEAI treatment is shown in Fig. 3b. It can be seen that the *J*_{sc} remains almost the same, while *V*_{oc} substantially increases compared with the control device. We also observed that the hysteresis can be eliminated when the perovskite surface is treated with PEAI. Both the improvement of device performance and elimination of hysteresis could be related to the reduction of the surface defects, which can result in suppression of charge recombination and also charge accumulation²⁰.

We tested the device performance when the PEAI layer was processed at different temperatures and found that the device performance degraded while thermal annealing the PEAI layer (Supplementary Fig. 11 and Supplementary Table 3). This is consistent with the PL results (Fig. 2a,b), confirming that the PEAI itself has a better passivation effect for perovskite layer. We also screened several organic iodine salts for perovskite surface treatment and found that the device performance either showed no significant enhancement or inferior performance compared with the control device (Supplementary Fig. 12 and Supplementary Table 4).

After optimization of the PEAI-treated device, we obtained the best device with an efficiency of 23.56%, a *V*_{oc} as high as 1.16 V, a *J*_{sc} of 24.9 mA cm^{−2} and a fill factor (FF) of 81.4% (Fig. 3c), and there is almost no hysteresis for the device (Supplementary Fig. 13 and Supplementary Table 5). The steady output for the best device shows a quasi-steady output of 23.2%, while for the device without PEAI, the output is 21.3% (Supplementary Fig. 14). The integrated current density from the external quantum efficiency (EQE) is 24.5 mA cm^{−2} (Fig. 3d), which is almost in agreement with the measurement from the solar simulator in this study. In addition, according to the EQE decrease to half the peak value and the differential of the EQE⁴⁰, it can be estimated that the absorption threshold is 1.53 eV. This value is consistent with the result from the differential of absorption (Supplementary Fig. 15). The statistics of the PCE distribution confirm the reproducibility of our results and that the PEAI treatment can undoubtedly improve the performance of PSCs (Supplementary Fig. 16).

We sent the best device to an accredited laboratory (Newport, USA) for certification. The devices showed 23.32% certificated efficiency with a *V*_{oc} of 1.18 V, *J*_{sc} of 25.2 mA cm^{−2} and FF of 78.4% (Supplementary Figs. 17 and 18). Furthermore, it is noted that the certificated efficiency obtained here is the quasi-steady-state efficiency⁴¹. In this new measurement standard, the *J*–*V* parameters were measured under a 13-point IV sweep configuration wherein the bias voltage (current for *V*_{oc} determination) is held constant until the measured current (voltage for *V*_{oc}) is determined to be unchanging at the 0.03% level. This is intended to represent the quasi-steady-state performance of the device.

We tested the device response under different light intensities (Supplementary Fig. 19). We found that both the control and PEAI-treated devices show a linear *J*_{sc} versus light intensity relationship (Supplementary Fig. 19c). This indicates that carriers can transport smoothly in the device and that there is no obvious charge barrier induced at the interface even though insulating PEAI was introduced. The relationship between *V*_{oc} and light intensity is also plotted (Fig. 4a). It can be seen that the control device shows a slope of $1.88k_B T/q$, while PEAI-treated devices show a much smaller slope

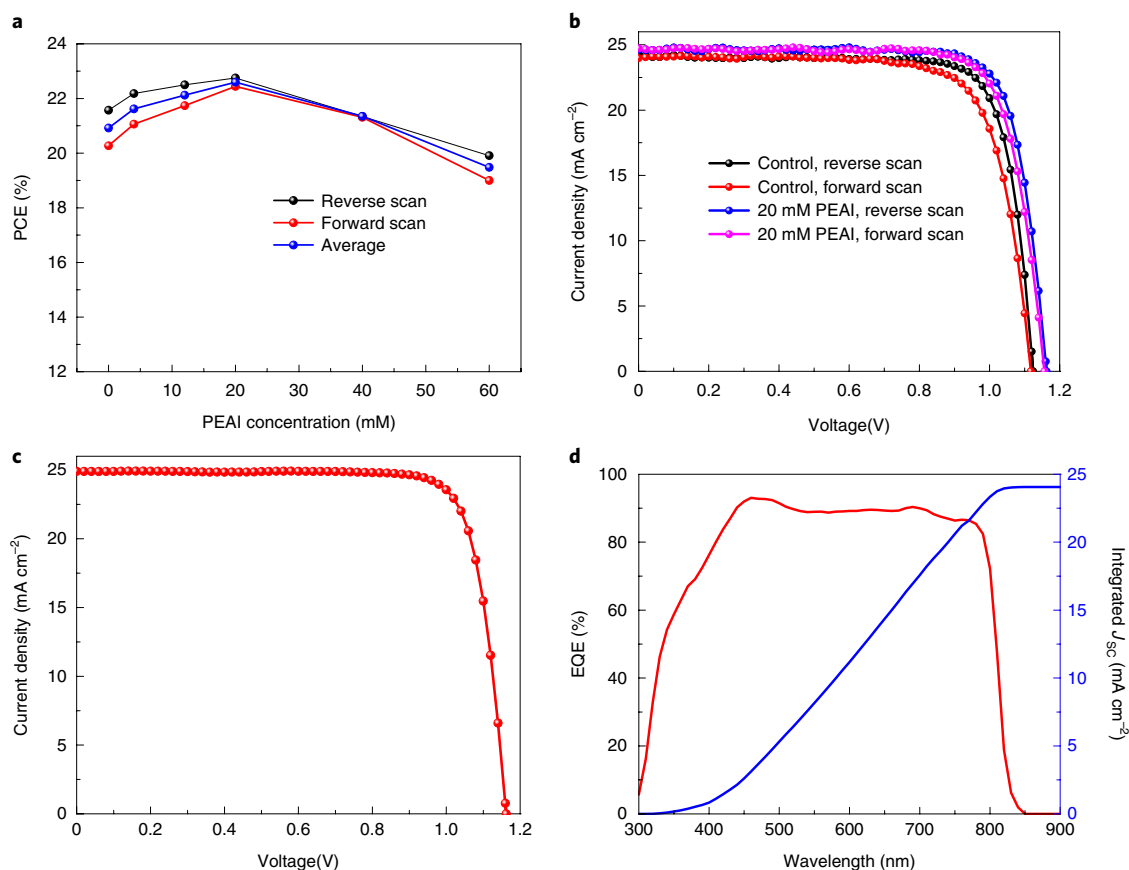


Fig. 3 | Device performance. **a**, The dependence of device performance on concentration of PEAI solution. Reverse scan (1.2 V to 0 V) and forward scan (0 V to 1.2 V) were both collected, and the average of the reverse and forward scans is included. **b**, The typical J - V curve of the device with PEAI (20 mM) and without PEAI treatment under one-sun (100 mW cm^{-2}) conditions. The reverse and forward scans are both included. **c**, The best device we achieved in our laboratory by optimizing the PEAI condition (20 mM). A 23.56% PCE is achieved with a V_{OC} of 1.16 V, J_{SC} of 24.9 mA cm^{-2} and FF of 81.4%. **d**, EQE of the best device. The integrated J_{SC} from the EQE is 24.5 mA cm^{-2} .

($1.27k_B T/q$), where k_B is the Boltzmann constant, T is temperature and q is the electric charge. It is known that deviation of the slope from $k_B T/q$ reflects defect-assisted recombination in the devices⁴². These results further confirmed that the recombination has largely been suppressed in the perovskite layer.

We derived the charge recombination and transport time constants (τ_r and τ_t , respectively) from the transient photovoltage and photocurrent decays measured at open and short circuit, respectively⁴³. The τ_r of the PEAI-treated device is as high as $138.2 \mu\text{s}$ (Fig. 4b), which is even longer than that of MAPbI₃ perovskite single-crystal devices under one-sun conditions⁴⁴. Furthermore, this value is substantially longer than that of the control device (versus $8.8 \mu\text{s}$, Fig. 4b), which could be attributed to slower surface charge recombination. We also found that the τ_t slightly decreases from $3 \mu\text{s}$ to $2.3 \mu\text{s}$ after PEAI treatment (Supplementary Fig. 20). The insertion of insulating PEAI cannot enhance the hole extraction from perovskite to spiro-OMeTAD and decrease τ_t . Therefore, the slight reduction of τ_t could also be due to the reduction of charge recombination on introduction of PEAI.

A V_{OC} as high as 1.18 V is obtained (Supplementary Figs. 17 and 18), which is 94.4% of the Shockley–Queisser limit V_{OC} (1.25 V) for the absorption threshold of 1.53 eV (ref. 45). In addition, there is only 0.35 V of V_{OC} loss ($E_{abs}/q - V_{OC}$) at the absorption threshold (E_{abs}) 1.53 eV, which is even less than that of crystal silicon solar cells (0.38 V) and comparable to GaAs solar cells (0.30 V)¹³. It has been shown that the EQE of the EL of the solar cells under bias voltage (operating as a light-emitting diode (LED)) can be used to estimate

the recombination in the devices⁴⁵. Higher radiative recombination emission efficiency could deliver high V_{OC} based on the equation:

$$V_{OC} = \frac{k_B T}{q} \ln \left(\text{EQE}_{EL} \frac{J_{ph}}{J_{em,0}} + 1 \right) \quad (1)$$

where EQE_{EL} is quantum efficiency when the injection current in the dark is equal to the photocurrent (J_{ph}) of the device under light illumination and $J_{em,0}$ is the current of re-emitted photons at room temperature⁴⁵.

We tested the solar cell device operating as a LED in the dark and under a voltage bias. We found that PEAI-treated devices showed obvious EL and an emission peak located at 805 nm (Fig. 4c). Furthermore, the device can even emit under lower applied voltage (1.2 V), which is lower than the absorption threshold of perovskite (1.53 eV)—the apparent violation of energy conservation could come from heat extracted from the diode⁴⁶. It is encouraging that the device showed close to 7% EL efficiency under an injection current of 25.2 mA cm^{-2} (equal to short-circuit current $J_{SC} = J_{ph}$) (Fig. 4d and Supplementary Fig. 21). For comparison, the control device only shows 0.8% EL efficiency under same injection current (Fig. 4d and Supplementary Fig. 21). Based on equation (1), it can be estimated that the V_{OC} improvement (ΔV_{OC}) after PEAI treatment is: $\Delta V_{OC} = k_B T/q \ln(\text{EQE}_{PEAI}/\text{EQE}_{\text{without PEAI}}) = 0.056 \text{ V}$. This result is almost consistent with the J - V results, while the control and PEAI-treated devices showed V_{OC} around 1.12 V and 1.18 V, respectively.

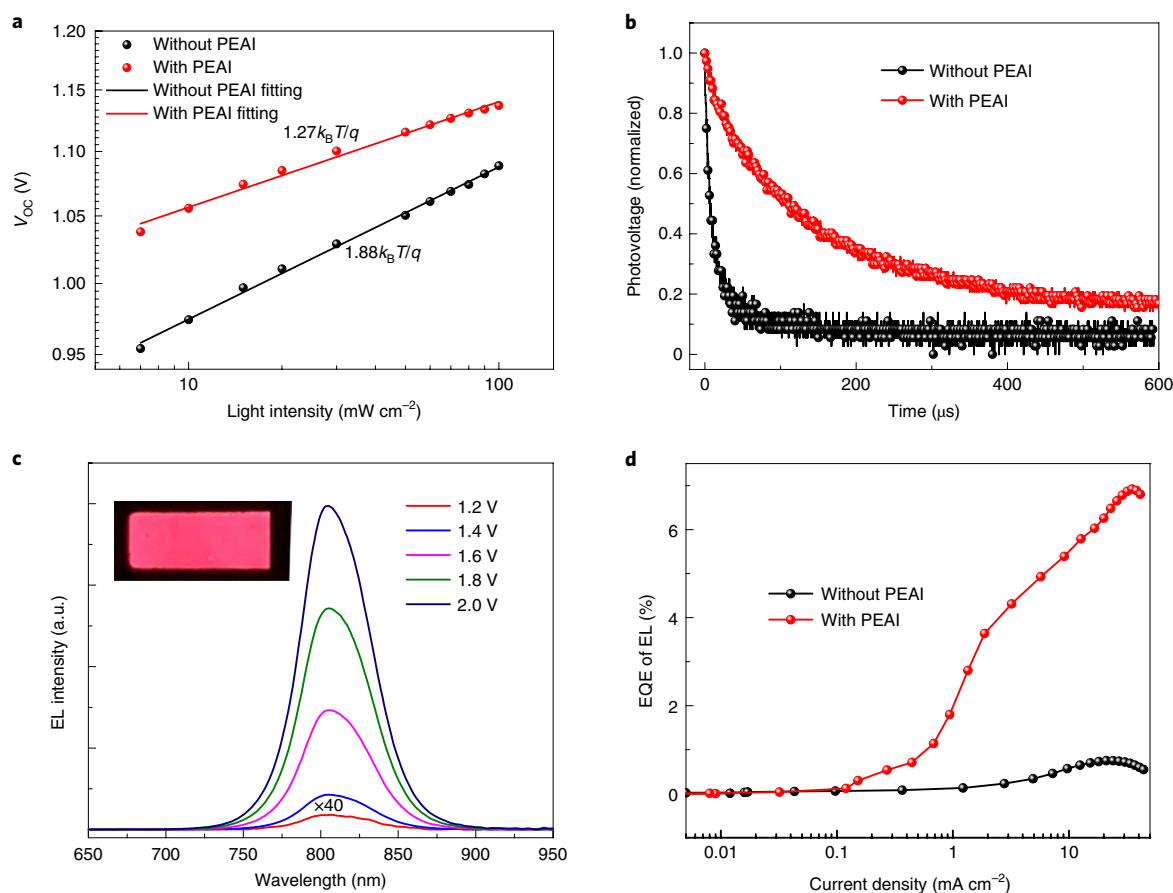


Fig. 4 | Characterization of the devices. **a**, V_{OC} versus light intensity for the PSCs with and without PEA. **b**, Normalized transient photovoltage decay curves for the solar cells with and without PEA. **c**, EL spectra of the devices with PEA under different voltage bias operating as LEDs. Inset: EL image of the devices under different voltage bias. **d**, EQE of EL of the devices while operating as LEDs.

We also calculated the internal quantum efficiency of the devices, which were 9.7% and 50.4% for the control and PEA-treated devices, respectively, according to the relationship between external and internal emission^{47,48}. This result shows that the non-radiative losses in the perovskite layer have been significantly suppressed by PEA passivation. We could increase the V_{OC} close to its limit (1.25 V) if we could fully suppress the non-radiative recombination in the perovskite layer, on the surface or at the interface, and then the PCE of PSCs could be improved beyond 25%.

Stability of the devices

We tested the device stability at 85°C for more than 500 h (Fig. 5). A more thermally stable hole transport layer poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) was used to replace spiro-OMeTAD for thermal stability testing^{49,50}. The initial PCE of the PEA-treated devices with a PTAA hole transport layer was 19.1% (Supplementary Fig. 22 and Supplementary Table 6), which is lower than the best device performance using spiro-OMeTAD as the hole transport layer. This is a common issue and could be addressed in future studies. We found that the PCE of the PEA-treated device decreased at the start of several hours of annealing at 85°C, and then the performance remained almost constant up to 500 h of heating. Time-resolved PL for the perovskite films/hole transport layer showed that the lifetime of the perovskite layer was significantly shortened in minutes after heating at 85°C (Supplementary Fig. 23). This is fully consistent with the results shown in Fig. 2b. Then there is no obvious change when the heating time is extended to hours (Supplementary Fig. 23). The shortening of lifetime during

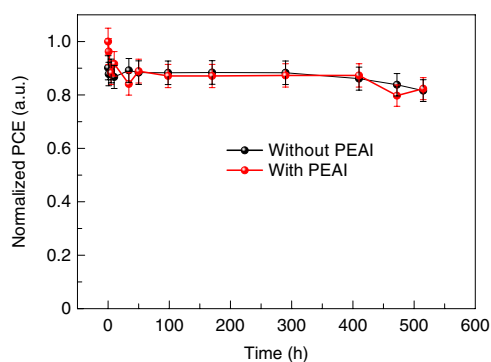


Fig. 5 | Thermal stability of the devices. The thermal stability (85°C) of the devices with and without PEA layers. Up to 500 h stability was tested. PTAA was used to replace spiro-OMeTAD as the hole transport layer for thermal testing. The device performance without PEA has been divided by the initial device performance of PEA-based devices, and the initial performance of the devices with PEA has been normalized. The starting efficiency of the cells is 19.1% and 17.2% for the devices with and without a PEA layer, respectively (Supplementary Fig. 22 and Supplementary Table 6). For each type of device, ten devices were utilized for the thermal stability test, 5% of error bars has been adopted.

heating could be ascribed to the disappearance of the passivation effect by converting PEA into PEA_2PbI_4 according to Fig. 2, which results in device performance degradation during heating.

These results showed that the thermal stability of PEAI-based devices is not good enough, and that other passivation materials with good thermal stability should be screened. We also preliminarily tested the device stability under continuous light soaking at the maximum power point (25°C, 100 mW cm⁻²) and found that there is no obvious decrease in device performance over 40 h (Supplementary Fig. 24).

In summary, we have provided an efficient and simple method for passivation of the perovskite surface through which the defects are significantly reduced and the recombination is suppressed. As a result, a 23.32% certificated efficiency was obtained by adopting a low-temperature solution-processed planar structure. This method could be helpful for obtaining low-cost, efficient and highly flexible solar cells.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of data availability and associated accession codes are available at <https://doi.org/10.1038/s41566-019-0398-2>.

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Author contributions

J.Y. conceived the idea, directed and supervised the project. Q.J. initialized this project, fabricated and characterized the devices. Y.Z. took part in the device fabrication and characterizations. X.Z., X.Y., C.Y., Z.C., Q.Y., X.L. and Z.Y. were involved in the data

analysis. J.Y., J.Q., Y.Z. and X.Z. co-wrote the manuscript. All authors contributed to discussions and to finalizing the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Methods

Materials. SnO₂ colloid precursor (tin (iv) oxide, 15% in H₂O colloidal dispersion) was purchased from Alfa Aesar. Dimethylformamide (DMF), dimethyl sulfoxide (DMSO) and lead iodine (PbI₂) were purchased from Sigma Aldrich. Formamidinium iodide (FAI), methylammonium bromide (MABr), methylammonium iodide (MAI), methylammonium chloride (MACl) and phenethylammonium iodide (PEAI) were all purchased from Xi'an Polymer Light Technology in China.

Solar cell fabrication. ITO glass was cleaned by sequentially washing with detergent, deionized water, acetone and isopropanol (IPA). Before use, the ITO was cleaned with ultraviolet ozone for 10 min. Then the substrate was spin-coated with a thin layer of SnO₂ nanoparticle film (2.67%, diluted by water) at 4,000 r.p.m. for 30 s, and annealed in ambient air at 150 °C for 30 min. It is better to clean the substrate with ultraviolet ozone for 10 min to improve the surface wetting. After that, 1.5 M of PbI₂ in DMF:DMSO (9:1) solvent was spin-coated onto SnO₂ at 1,500 r.p.m. for 30 s, then annealed at 70 °C for 1 min, and then cooled to room temperature. For FAMAPbI₃ perovskite film deposition, a solution of FAI:MAI:MACl (90 mg:6.39 mg:9 mg in 1 ml IPA) was spin-coated onto the PbI₂ at spin rate of 2,000 r.p.m. for 30 s, followed by thermal annealing at 150 °C for 15 min in ambient air conditions (30–40% humidity). For comparison, a (FAPbI₃)_{1-x}(MAPbBr₃)_x mixed perovskite were also synthesized. A solution of FAI:MABr:MACl (90 mg:9 mg:9 mg in 1 ml IPA) was spin-coated onto the PbI₂ at spin rate of 2,000 r.p.m. for 30 s, followed by thermal annealing at 150 °C for 15 min in ambient air conditions (30–40% humidity). After perovskite formation, the samples were transferred to a nitrogen-filled glove box for further processing. For PEAi treatment, the PEAi solution was dissolved in IPA with different concentrations ranging from 0 mM to 60 mM and spin-coated onto the perovskite surface at a spin rate of 5,000 r.p.m. without any further processing. Then the hole transporting layer was deposited on top of the perovskite layer at a spin rate of 3,000 r.p.m. for 30 s using spiro-OMeTAD solution, which consisted of 72.3 mg spiro-OMeTAD, 35 µl bis(trifluoromethane) sulfonimide lithium salt (LiTFSI) stock solution (260 mg LiTFSI in 1 ml acetonitrile), 30 µl 4-tertbutylpyridine and 1 ml chlorobenzene. For the device thermal stability test, PTAA doped with poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (TPFB) was used to replace spiro-OMeTAD as the hole transport layer. The concentration of PTAA was 30 mg ml⁻¹ and the weight ratio of PTAA/TPFB was 10:1 (ref. ³⁰). The PTAA was deposited on top of the perovskite layer at a spin rate of 2,000 r.p.m. for 30 s. Finally, 80 nm of Au film was thermally evaporated as a counter electrode using a shadow mask. The device size areas were 0.108 cm². When measuring, a 0.0739 cm² non-reflective mask was used to define the accurate active cell area.

Material and device characterization. The scanning electron microscopy images was acquired using a field-emission scanning electron microscopy (FEI NanoSEM650), which uses an electron beam accelerated at 500 V to 30 kV, enabling operation at a variety of currents. Atomic force microscopy measurements were carried out using a Solver P47 PRO. The XRD patterns of the perovskite thin films were obtained on a Rigaku D/MAX-2500 system using Cu K α radiation ($\lambda = 1.5405$ Å) as the X-ray source. Scans were taken with a 0.5-mm-wide source and detector slits, with X-ray generator settings at 40 kV and 30 mA. For conventional XRD, a θ – 2θ model was used. For the GIXRD measurement, the incident angle was 0.5°. Absorption spectra were obtained using an ultraviolet–visible spectrometer (Cary 5000). Steady-state PL was measured by a FLS980 spectrometer. Time-resolved PL spectra were obtained using a F900 spectrometer. XPS was performed on a Thermo Scientific ESCALab 250Xi using 200 W monochromated Al K α (1486.6 eV) radiation. A 500 µm X-ray spot was used for XPS analysis. The base pressure in the analysis chamber was about 3×10^{-10} mbar. For the PL mapping, a laser scanning confocal microscope (Enlitech, SPCM-1000) equipped with a 470 pulse laser and galvo-based scanner was used. The objective was $\times 50$ with numerical aperture of 0.8; for this objective, the confocal pinhole was set to 50 µm size. The detector was a photo-multiplier tube (Hamamatsu) with a photon-counting module. The laser source was set to 1 µW range on the sample for measurement. Transient photocurrent and photovoltage measurements were measured on a home-made system similar to that previously reported⁸. For the device operating as a LED, the device was mounted in an integrating sphere and the power output measured for calculation of the EQE (Enlitech, LQE-50-PL). The J – V characteristics of the photovoltaic cells were obtained using a Keithley 2400 Source Meter under simulated one-sun AM 1.5G illumination (100 mW cm⁻²) with a solar simulator (Enlitech, SS-F5-3A) and the light intensity was calibrated by means of a KG-5 Si diode. The J – V measurements were carried out in ambient air. The devices were measured both in reverse scan (1.2 V $\rightarrow$ 0 V, step 0.02 V) and forward scan (0 V $\rightarrow$ 1.2 V, step 0.02 V). The EQE was measured using an Enlitech EQE measurement system (QE-R3018). We sent the best devices to an accredited laboratory (Newport, USA) for certification. The device thermal stability tests were carried out in a nitrogen glove box, and the devices were put on a hotplate with a temperature of 85 °C.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

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► Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

Area of the tested solar cells	☒ Yes	Methods in the manuscript
	☐ No	
Method used to determine the device area	☒ Yes	Methods in the manuscript
	☐ No	

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction	☒ Yes	Fig.3b
	☐ No	
Voltage scan conditions <i>For instance: scan direction, speed, dwell times</i>	☒ Yes	Methods in main text
	☐ No	
Test environment <i>For instance: characterization temperature, in air or in glove box</i>	☒ Yes	Methods in main text
	☐ No	
Protocol for preconditioning of the device before its characterization	☒ Yes	Methods in main text
	☐ No	
Stability of the J-V characteristic <i>Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see ref. 7 for details.</i>	☒ Yes	Supplementary Fig. 17, Newport certificated results
	☐ No	

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization	☐ Yes	Not relevant
	☒ No	
Related experimental data	☐ Yes	Not relevant
	☒ No	

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)	☒ Yes	Fig. 3d
	☐ No	
A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes	In the main text
	☐ No	
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☐ Yes	State where this information can be found in the text.
	☐ No	Explain why this information is not reported/not relevant.

5. Calibration

Light source and reference cell or sensor used for the characterization	☒ Yes	Methods in the main text
	☐ No	
Confirmation that the reference cell was calibrated and certified	☒ Yes	Methods in the main text
	☐ No	

Calculation of spectral mismatch between the reference cell and the devices under test

☒ Yes
☐ No

It has been calibrated

6. Mask/aperture

Size of the mask/aperture used during testing

☒ Yes
☐ No

Methods in the main text

Variation of the measured short-circuit current density with the mask/aperture area

☒ Yes
☐ No

Newport certification report in supplementary Fig. 17

7. Performance certification

Identity of the independent certification laboratory that confirmed the photovoltaic performance

☒ Yes
☐ No

Newport

A copy of any certificate(s)

Provide in Supplementary Information

☒ Yes
☐ No

Supplementary Fig. 17 and Fig. 18

8. Statistics

Number of solar cells tested

☒ Yes
☐ No

Supplementary Fig. 16

Statistical analysis of the device performance

☒ Yes
☐ No

Supplementary Fig. 16

9. Long-term stability analysis

Type of analysis, bias conditions and environmental conditions

For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature

☒ Yes
☐ No

Fig. 5 and Supplementary Fig. 24